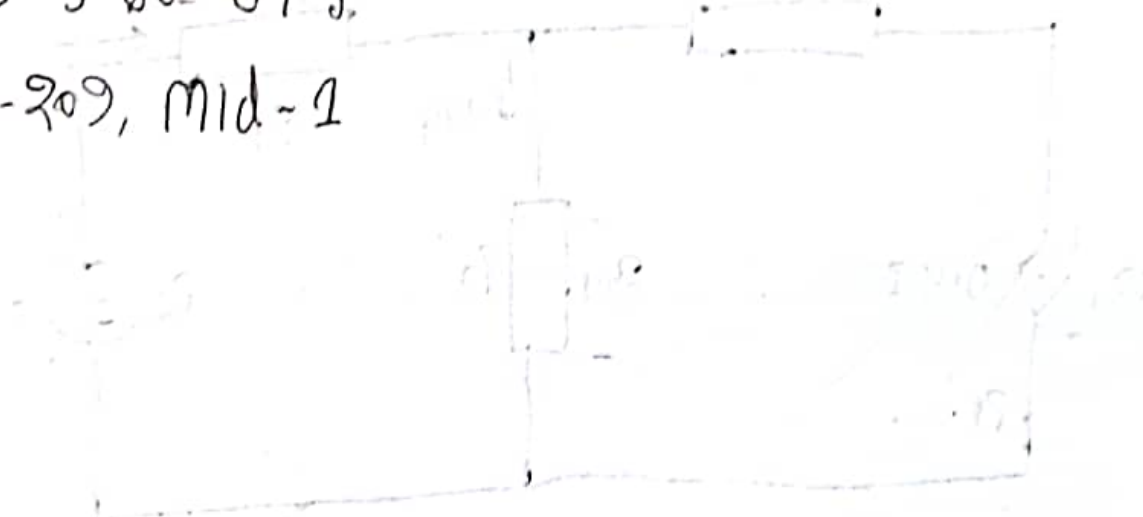


Oli Ahmed

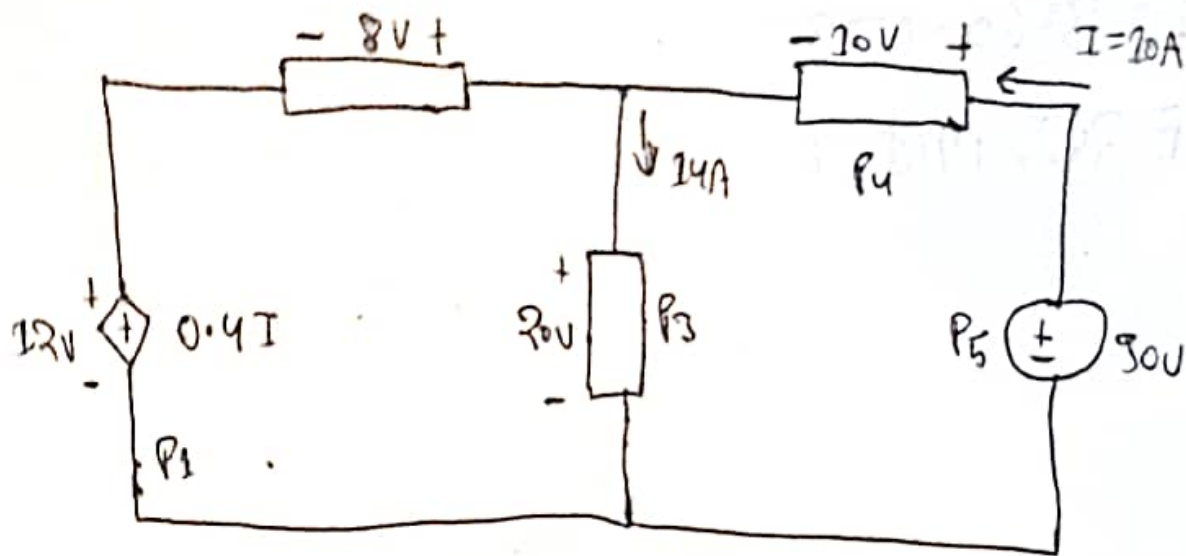
2019-3-60-073

CSE-209, Mid-1



Handwritten notes in the bottom right corner, including the expression $(10 \times 10^3) - 0$ and other illegible scribbles.

Ans to the ques No-1



We know,

$$P = VI$$

Hence,

$$\begin{aligned} P_1 &= -(12 \times 0.4I) \\ &= -(12 \times (0.4 \times 10)) \\ &= -48 \text{ watt} \end{aligned}$$

(supply)

$$P_2 = -(8 \times 4)$$
$$= -32 \text{ watt}$$

(supply)

$$P_3 = 20 \times 14$$
$$= 280 \text{ watt}$$

(absorve)

$$P_4 = 10 \times 10$$
$$= 100 \text{ watt}$$

(absorve)

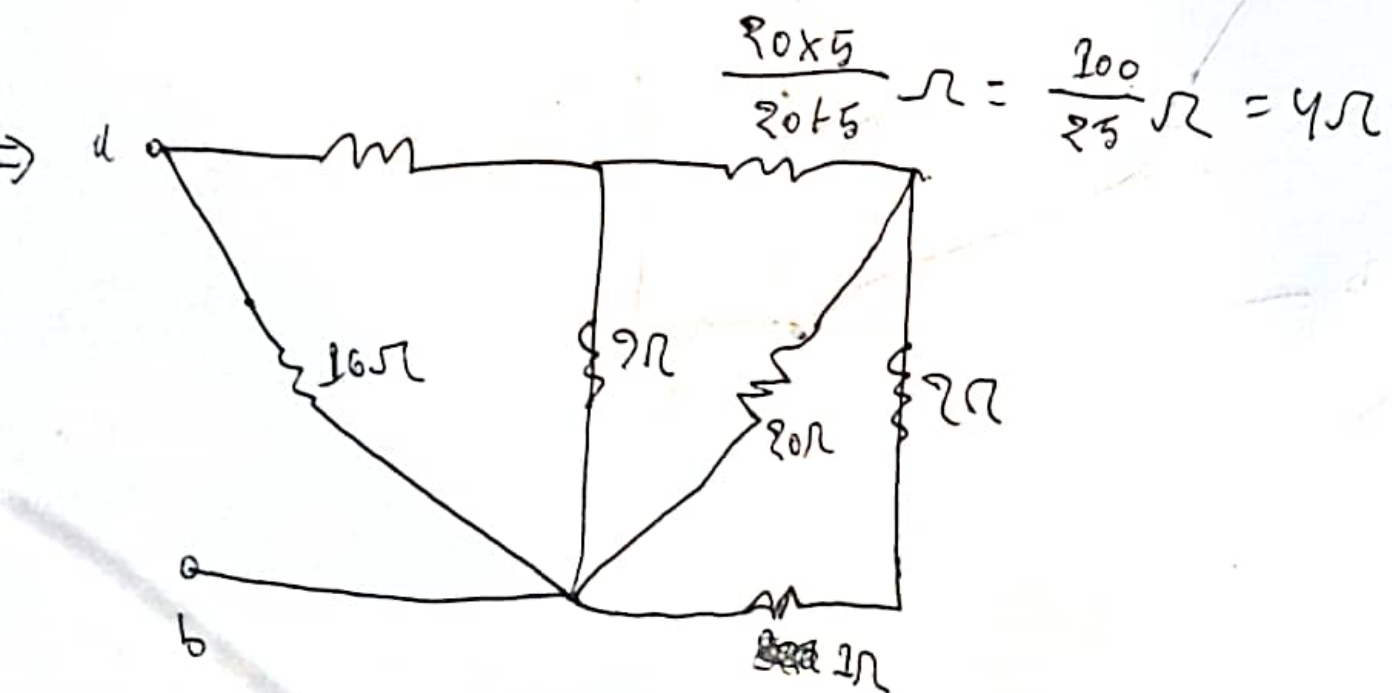
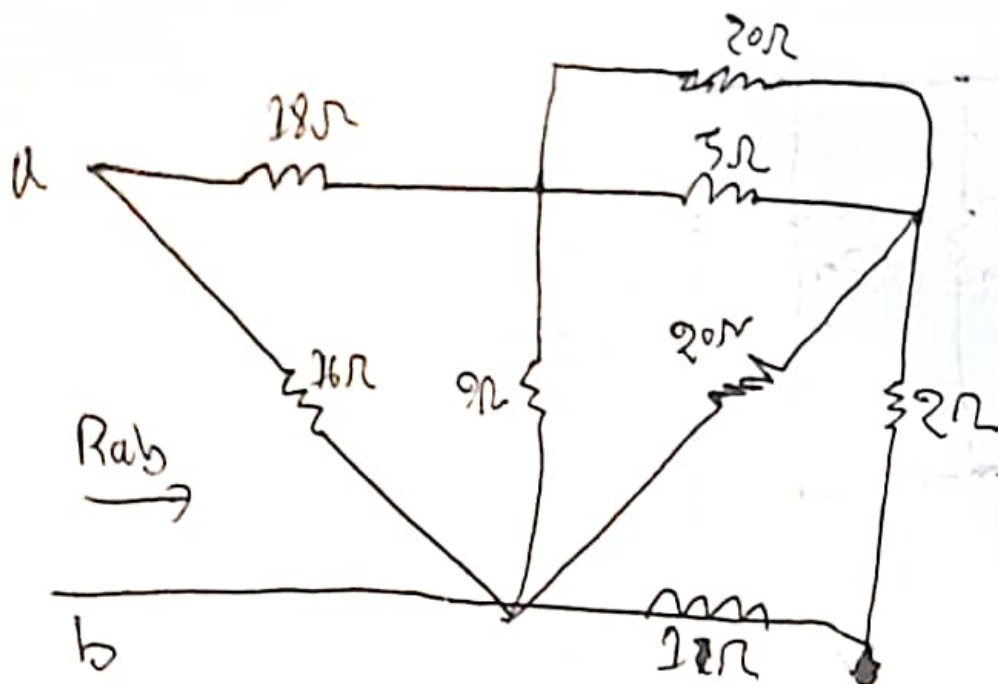
$$P_5 = -(30 \times 10)$$
$$= -300 \text{ watt}$$

(supply)

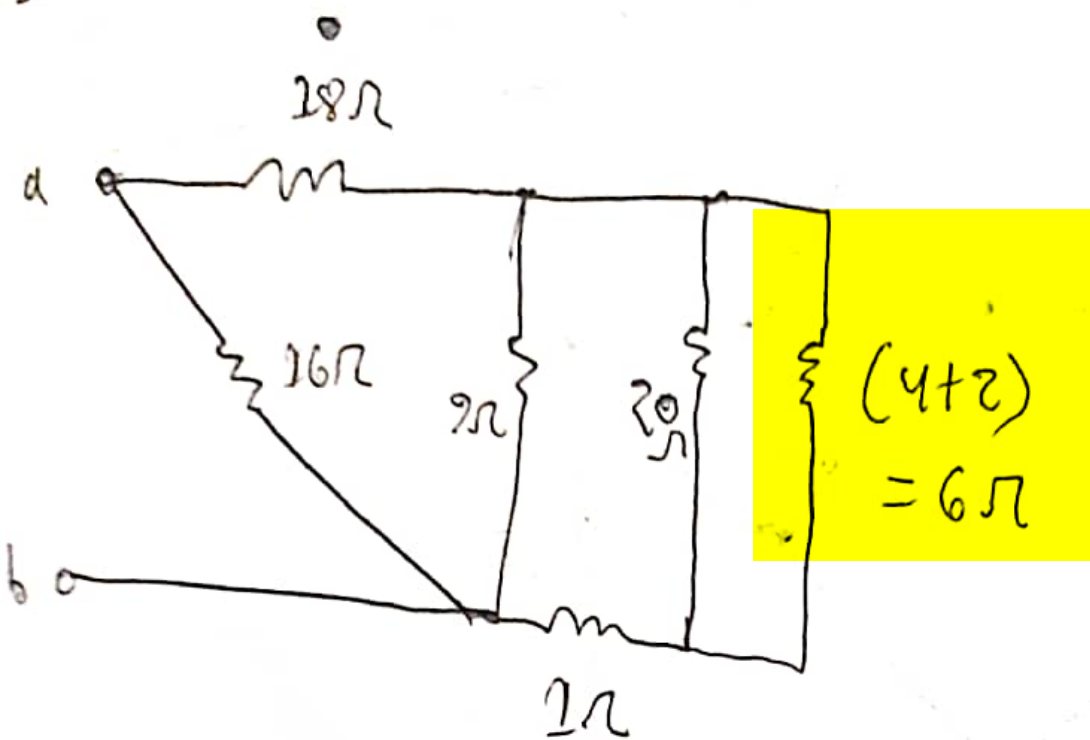
Do

~~Ans~~

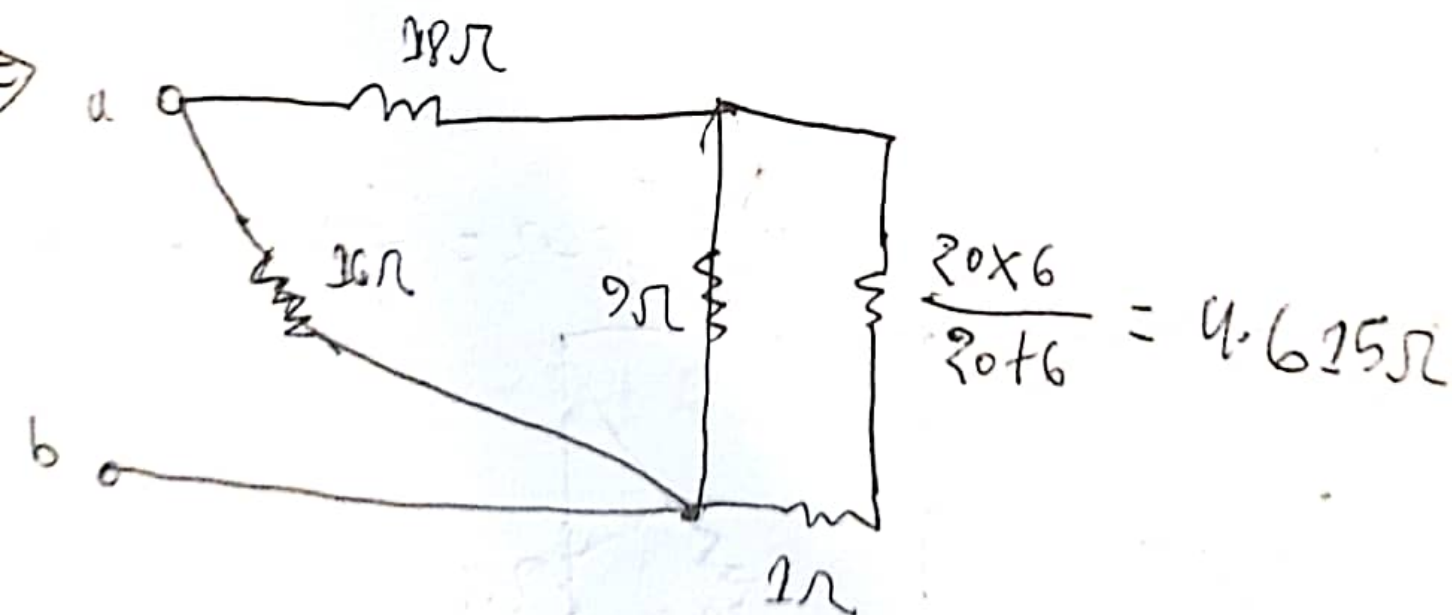
Ans to the ques no-2

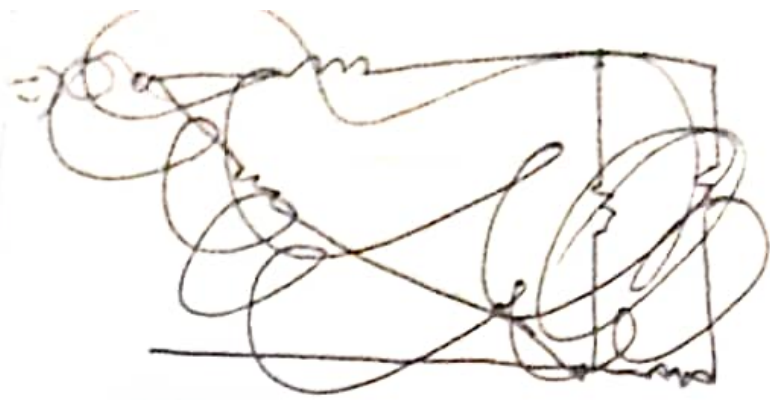


⇒

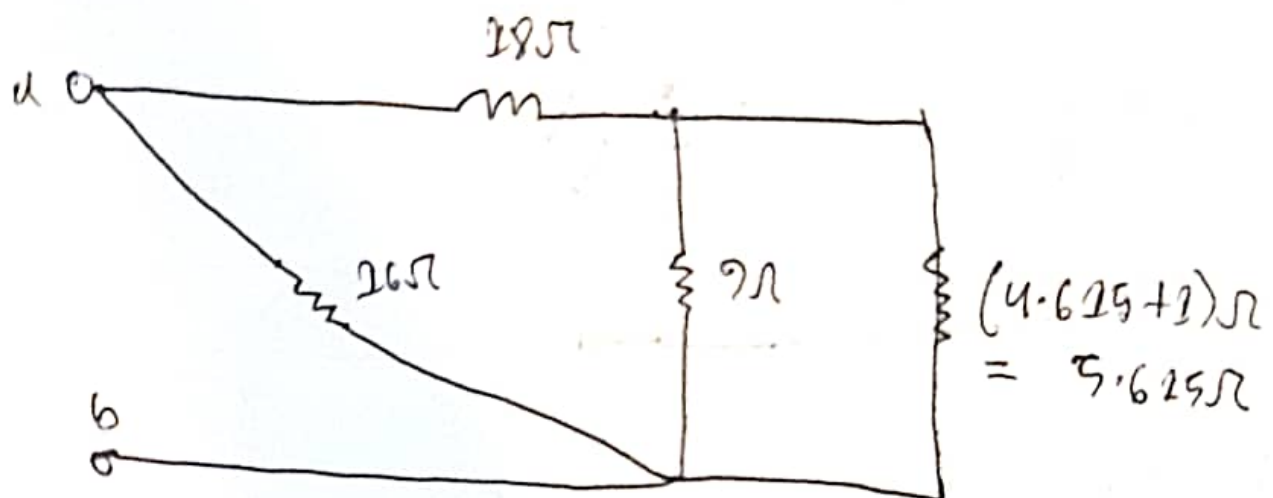


⇒

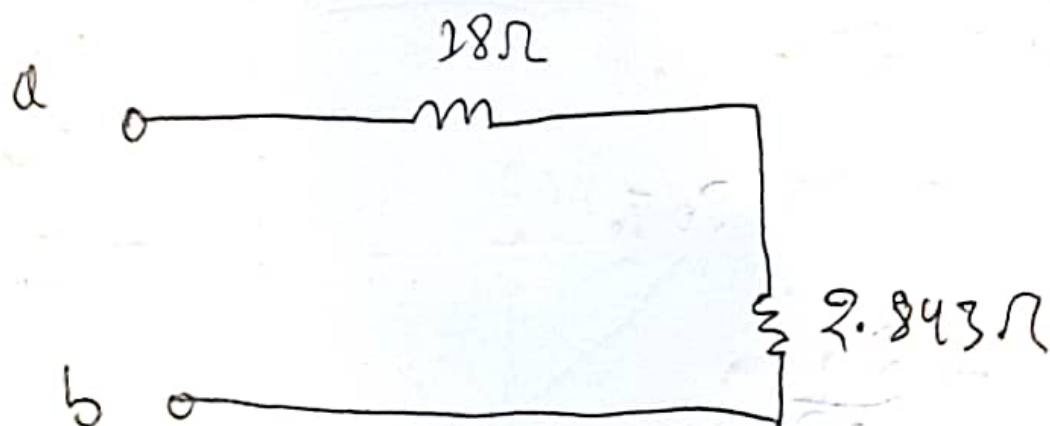




⇒



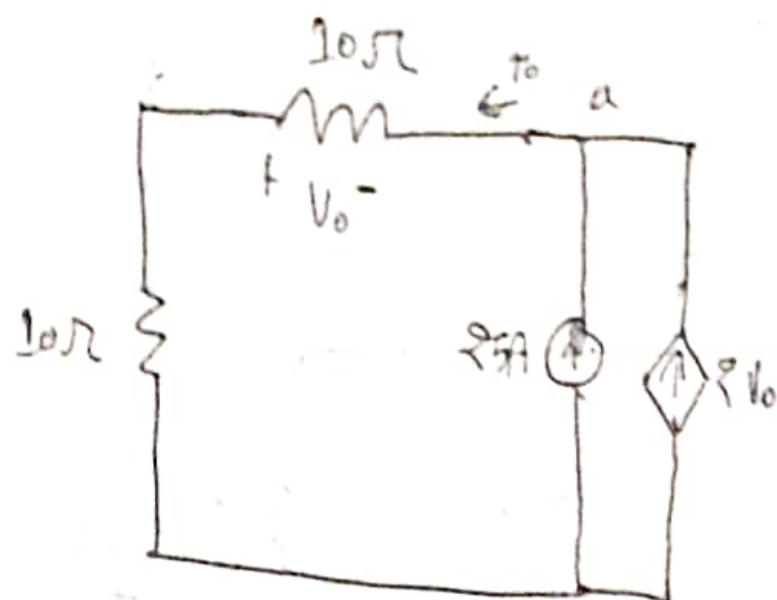
⇒



$$\therefore R_b = (18 + 2.843)\Omega = 20.843\Omega$$

Ans.

Ans. to the ques no-3



Applying KCL at node a,

$$\begin{aligned} I_o &= 25 + 2V_o \\ &= 25 + 20I_o \end{aligned}$$

$$I_o = -\frac{25}{19}$$

$$I_o = -1.31$$

Hence,

$$V_o = I_o R$$

$$= I_o \times 20$$

$$\therefore V_0 = I_0 R$$

$$= -1.81 \times 10$$

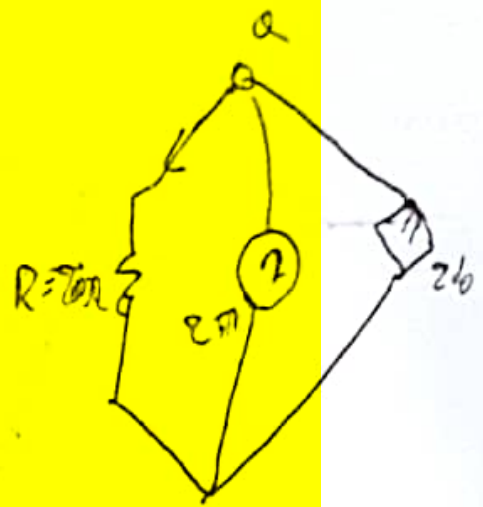
$$= -18.1 \text{ V}$$

Hence,

$$V = I_0 \times R$$

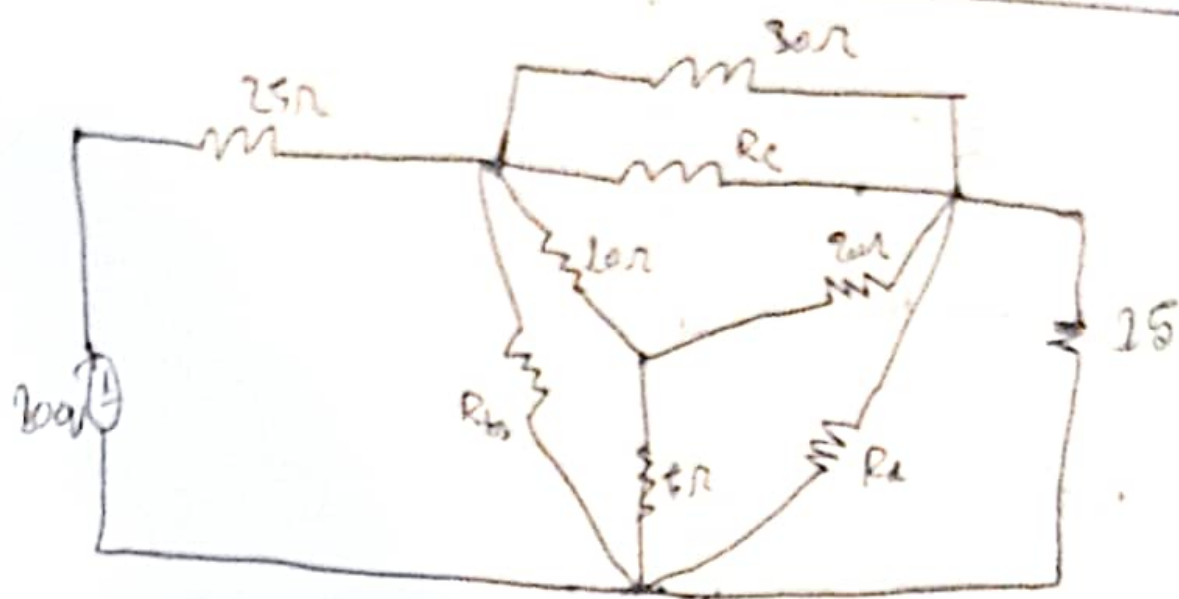
$$= 1.3 \times 20$$

$$= -26.5$$



Q.1

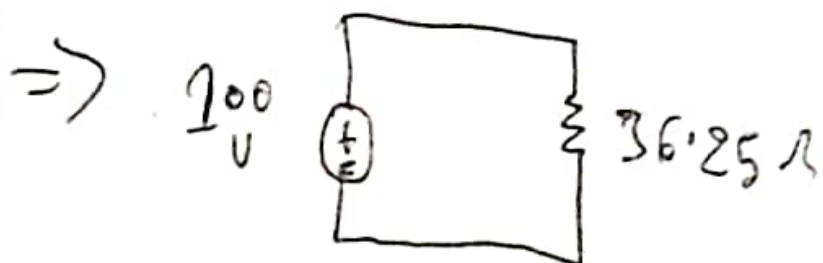
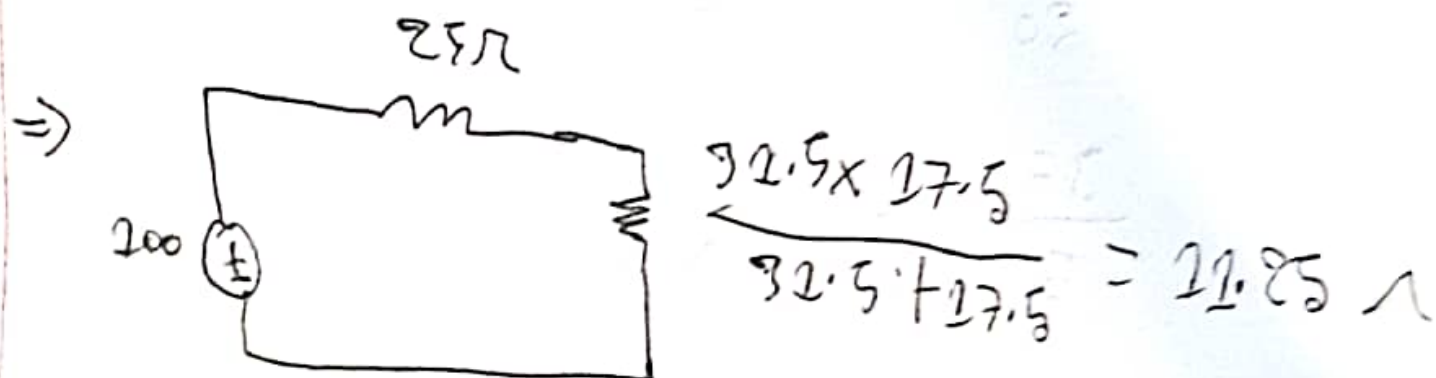
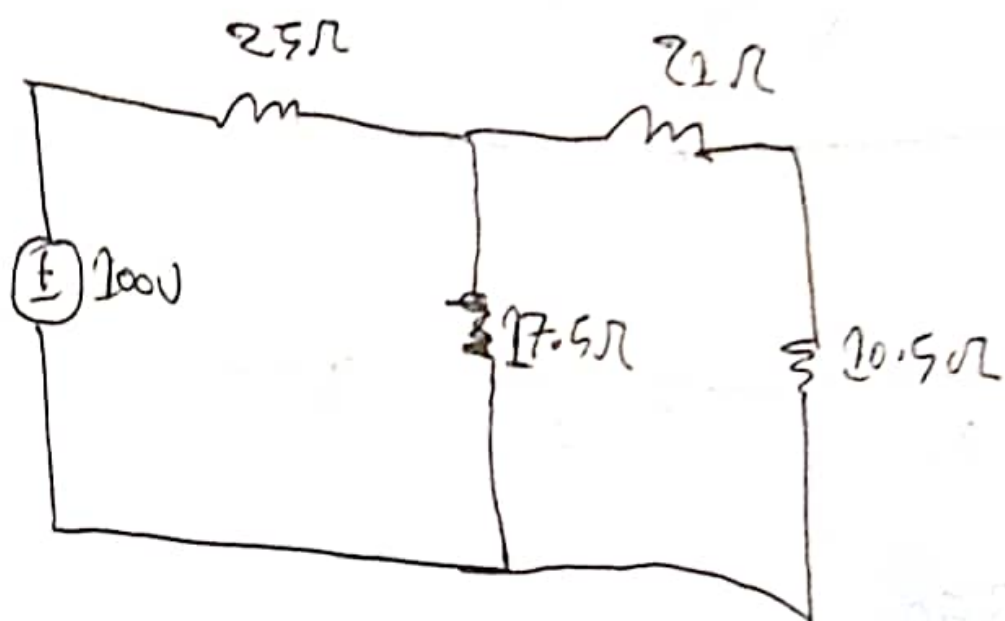
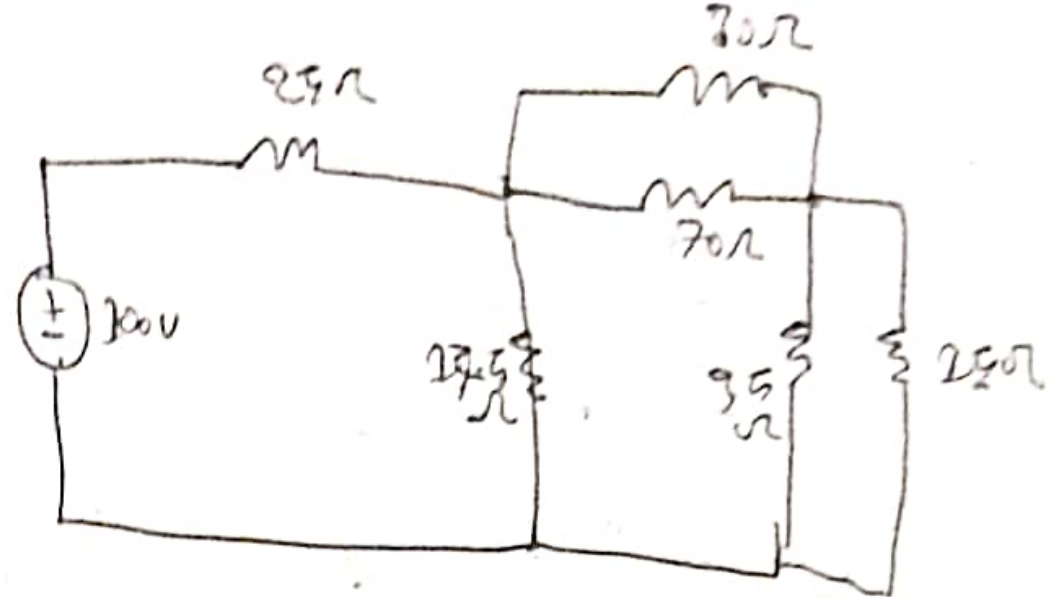
Ans. to the ques - 4



$$R_a = \frac{200 + 100 + 50}{10} = \frac{350}{10} = 35 \Omega$$

$$R_b = \frac{350}{20} = 17.5 \Omega$$

$$R_c = \frac{350}{5} = 70 \Omega$$



Now,

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{100}{36.25} \\ &= 2.75 \text{ A} \end{aligned}$$

Ans